

Research Statement

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As an algebraic geometer, I use homological algebra as an interface between algebraic geometry and representation theory. I am especially interested in studying questions using structures from various areas different from its familiar formulation. This viewpoint has many successful precedents in modern mathematics, such as Hodge theory [86], Tannakian reconstruction [30], and various reconstruction theorems in algebraic geometry [34, 14, 8]. In my work, mathematical objects are often viewed in different settings where new structures become available and helpful.

Recently, my work concentrate mostly around finite-dimensional algebras and DG algebras. For example: Han's conjecture, as a classical representation theory question, is settled by transporting a nontrivial input from algebraic geometry [48]; phantom categories, first discovered in algebraic geometry, admits a much smaller algebraic realization on a 3-vertex quiver [65], by observing a similar structure in a classical quiver example; Cartan determinant, a numerical invariant originating in finite-dimensional algebra, extends to algebraic geometry, and Hodge-theoretic tools forces positivity in a broad range of cases [66]. Below I elaborate my research in more detail.

Homological algebra as a meeting point

Finite-dimensional algebras are central objects in representation theory. Their module categories can be studied by quivers with relations. This makes them both flexible enough to encode representation-theoretic behaviors, and concrete enough for explicit computations [6, 5]. A natural intrinsic homological property of an algebra is smoothness. Geometrically, this notion corresponds to smoothness of a projective variety.

It is natural to ask if smoothness can be detected by computable invariants. Hochschild homology is a natural candidate: it is a derived invariant and behaves well under many categorical operations. This question has a long history. In 1989, [40] asked whether eventual vanishing of Hochschild *cohomology* implies smoothness. Later [20] found counterexamples. In 2006, [39] proposed the homological version, known as the *Han's conjecture*. It was verified for many important classes, including commutative algebras, monomial algebras, and several classes preserved by standard constructions [39, 11]. However, the general formulation remained open for 2 decades. In [48], we show that Han's conjecture fails. We construct a finite-dimensional $\mathbb{C}$ -algebra A such that

$$\text{gldim}(A) = \infty, \quad \text{HH}_n(A) = 0 \quad \text{for every } n \geq 1.$$

Thus by Hochschild homology we cannot distinguish it from a smooth algebra. This result grew out of our earlier DG counterexample [64], where we first showed that finite Hochschild homology can coexist with nonsmoothness. The remaining problem was to realize the same phenomenon for an ordinary finite-dimensional algebra.

Essentially, the origin of this phenomenon is a categorical construction. The singularity is carried by a nonzero category with vanishing Hochschild homology. More precisely, the input is a *phantom category*: a nonzero smooth and proper category with vanishing Grothendieck group and Hochschild homology. We transport Krah's phantom category on rational surfaces [49] into the singularity category of an ordinary finite-dimensional algebra. This connection also brings my interests back to phantom categories themselves.

Phantom categories first appeared from surfaces of general type [36, 12]. Krah's striking construction occurs on rational surfaces [49], and recent work has produced infinite families on the same rational surfaces [67]. These beautiful geometric constructions are very useful for existence questions. However, the residual category itself is often difficult to study directly, since the ambient geometric category is relatively complicated.

This motivated us to look for phantoms inside much smaller algebraic models. In [65], we use a completely different method from [67] to produce infinitely many pairwise non-derived-Morita-equivalent phantom categories, with distinct Hochschild cohomology groups. The ambient categories are 25-dimensional path algebras of 3-vertex directed quivers with relations. Our inspiration comes from a classical example in representation theory [15, 52], and the construction also suggests a general method to produce more residual categories.

The small size of these examples is important. The ambient algebra, exceptional modules, residual category, and Hochschild cohomology can all be studied explicitly. Questions that are difficult for geometric phantoms, such as t -structures, moduli of objects, minimal A_∞ -models, and finite-cell presentations, now become accessible to direct calculation. Moreover, these new phantom categories can also be applied to our method disproving Han's conjecture. This produces smaller counterexamples whose internal structures can be observed more directly.

Another direction comes from the Cartan determinant. For a smooth finite-dimensional algebra, its Cartan matrix M_χ is unimodular. The classical Cartan determinant conjecture predicts $\det(M_\chi) = 1$ [32, 88]. For a smooth proper DG category $\mathcal{A}$, Euler pairing on the numerical Grothendieck group $K_{\text{num}}(\mathcal{A})$ gives a natural generalization. There, we may not have $\det(M_\chi) \in \{\pm 1\}$, but it appears that for every known smooth proper $\mathcal{A}$ with $\text{rk} K_{\text{num}}(\mathcal{A}) > 0$ in the literature, we have $\det(M_\chi) > 0$. So a natural conjecture is if $\det(M_\chi) > 0$ for any smooth and proper DG category. Geometry gives strong evidence for this positivity. In [66], we prove positivity for smooth projective varieties in odd dimension, and in even dimension when the numerical Hodge standard conjecture holds. The proof uses Hirzebruch–Riemann–Roch to translate the Euler form into intersection pairings on algebraic cycles. The sign is controlled by positive/negative definiteness in Hodge factors. This is another example where a numerical problem coming from finite-dimensional algebra becomes more rigid after it is placed in a geometric setting. It is known that the conjecture holds when the underlying algebra is graded, and our proof uses a genuinely different structure. My current work on explicit finite-cell DG algebras is also motivated by a different perspective of smooth and proper DG algebra from familiar construction. In representation theory, we often first fix a finite dimensional algebra and study smoothness. However, finite-cell presentations have smoothness automatically, while properness is the more interesting question. What this brings is rich combinatorial structures. I am looking for concrete finite criteria that guarantee properness. Such models may provide tests for the Cartan determinant problem, and more importantly, new concrete smooth proper category examples different from the algebro-geometric constructions.

Interactions with algebraic geometry

Following the above philosophy, geometry is a powerful additional structure that provides obstructions difficult to see from homological algebra alone. An example is classifying exceptional bundles on $\mathbb{P}^3$. Unlike on $\mathbb{P}^2$, it is still unknown if every exceptional vector bundle on $\mathbb{P}^3$ can be obtained by mutations from the standard exceptional collection. I showed that there is no exceptional vector bundle on $\mathbb{P}^3$ whose rank is maximal possible, in the sense of only using homological constraints [61]. The question is eventually reduced to studying certain cohomology groups on pencils of K3 surfaces, where a general algorithm was developed in my previous work [58].

This example is close to my current DG work. In a different direction from constructing phantoms [65], I am also interested in finding rank-1 residuals *without* exceptional objects. Choosing another suitable 3-vertex ambient quiver, I obtained such categories which does not possess any obvious exceptional objects (In particular, it is not equivalent to $\text{Perf}(k)$). However, it turned out that excluding exceptional objects rigorously is extremely difficult, because we no longer have abundant geometric objects to restrict to as in [61].

The interchange between algebra and geometry also works in the other direction. Not only geometric structures provide information unavailable in pure algebra, but also homological algebra can reveal new features of geometric objects. My work on stacks and rigid vector bundles develops the second direction.

Algebraic stacks combine geometry with representation theory in a natural way. Stabilizer groups contribute representation-theoretic data that is absent from the coarse space. For theories of Delign–Mumford stacks, root stacks are fundamental examples to test first. On surface root stacks, we construct (tilt) Bridgeland stability conditions, and show the support property with respect to rational Chen–Ruan cohomology where the stacky data are recorded [63]. Our construction is closely related to classical moduli theory. We prove a large-volume-limit theorem, locating a twisted semistability chamber, in the sense of [70]. This provides tools to study moduli theory of parabolic sheaves, and eventually moduli theory on maximal orders, where compactifications and deformation theory are known in important cases but the general moduli theory remains much less developed than for ordinary sheaves [50].

To understand derived categories of maximal orders as twisted root stacks, we study that of root gerbes as a starting point. Twisted sheaves have a well-developed Morita theory on schemes [56], while on stacks this is a new subject. Using homological algebra on Azumaya algebras, we obtain a classification there [62]. In particular, a Brauer class does not determine the Morita class, and Brauer twisting can change whether a category decomposes.

The statement develops 4 directions: finite-dimensional and DG algebras; phantom categories; derived categories of stacks; and vector bundles in classical algebraic geometry. The first 2 are the current center of my work. The latter 2 remain important directions and continue to provide structures and motivations for the homological questions.

1. Finite-dimensional algebras and DG algebras

Finite-dimensional algebras are basic objects in representation theory. Their module categories encode systems of linear maps and bring together quivers, homological algebra, and categorical methods [6, 5]. DG algebras keep many of these explicit features while allowing derived information to be built into the algebra itself. Quivers with relations, smoothness, Hochschild (co)homology, and Cartan matrix can all be studied in this setting. Derived Morita theory also keeps these algebraic models connected to algebraic geometry [46, 83].

1.1. Han’s conjecture for finite-dimensional algebras

Smoothness is one of the most basic homological properties of an algebra. For finite-dimensional algebras over perfect fields, it is equivalent to finite global dimension. In particular, smoothness implies finiteness of Hochschild homology. Han’s conjecture asks the converse: Does finiteness of Hochschild homology imply smoothness?

This question has a long story from history. In 1989, Happel asked whether eventual vanishing of Hochschild cohomology forces finite global dimension [40]. Later, [20] disproved this statement. Then in 2006, Han proposed the homological version [39]. Unlike the cohomological conjecture, Han’s conjecture remained open for 2 decades. It was proved for many families, including commutative algebras, monomial algebras, and several classes stable under standard constructions [39, 11]. Our counterexample [48] settled the conjecture negatively, even when all positive Hochschild homology vanishes.

Theorem 1.1 ([48]). *There exists a finite-dimensional $\mathbb{C}$ -algebra A with infinite global dimension such that*

$$\mathrm{HH}_n(A) = 0 \quad \text{for every } n \geq 1.$$

The main input comes from algebraic geometry. Krah’s universal phantom on $\mathrm{bl}_{10}\mathbb{P}^2$ is nonzero with vanishing Hochschild homology [49]. The central technical bridge is Chen’s partial-resolution theorem [28], which allows us to pass to an ordinary finite-dimensional algebra whose singularity category is related to this phantom. The relation is made precise by 1-periodic folding construction. The nonzero phantom with $\mathrm{HH}_* = 0$ is eventually turned into a nonzero singularity category with $\mathrm{HH}_* = 0$, which produces non-smoothness of our algebra with $\mathrm{HH}_{>0} = 0$. Thus the counterexample comes from transporting a geometric residual category into a classical algebraic setting, rather than from a direct calculation on the final algebra.

The three-vertex phantom construction gives a useful test of how much information the Han construction forgets. In [65], we construct pairwise non-derived-Morita-equivalent phantom categories $\mathcal{P}_m$ by regrading one common finite-dimensional bigraded model. In current calculations, all $\mathcal{P}_m$ have equivalent one-periodic foldings. Since the later steps of the Han construction only use this folded category, these different phantom inputs lead to the same ordinary algebra. Thus one Han counterexample can have many pairwise inequivalent phantom antecedents: the choice of the starting phantom is highly non-unique.

From another perspective, our new phantom categories allow us to replace the large geometric phantom by a much smaller model and observe its internal structure more closely.

Question 1.2. *Apply our construction above to our 3-vertex phantom model in [65]. The goal is to obtain a much smaller ordinary counterexample for which the quiver, relations, singularity category, and finer Hochschild invariants can all be computed directly.*

The following research programs fall into the general theme “what numerical cohomological invariants detect smoothness”. First, we observe that the counterexample of Happel’s HH^* question has infinite HH_* , and our counterexample of Han’s HH_* question has infinite HH^* . Since smoothness implies finiteness of both HH_* and HH^* , we have the following natural question.

Question 1.3. *Can Happel’s and Han’s detection principles fail simultaneously? More precisely, is there a non-smooth finite-dimensional algebra A where both $\mathrm{HH}_*(A)$ and $\mathrm{HH}^*(A)$ are finite-dimensional?*

More generally, recently [4] introduced the *Serre–Hochschild plane* following [57]:

$$\mathbb{T}^{p,m}(A), \quad (p, m) \in \mathbb{Z}^2,$$

for every finite-dimensional algebra A . It packages Hochschild cohomology with coefficients in derived powers of the Serre bimodule $\omega = DA$. The column $m = 0$ contains Hochschild cohomology, while the column $m = 1$ is dual Hochschild homology; the subsequent columns contain further Hochschild-type invariants. Importantly, the plane is defined without assuming that ω is invertible. Thus Han's conjecture asks whether finiteness of one distinguished column of a much larger invariant can detect finite global dimension. This leads to a hierarchy of smoothness-detection problems.

Program 1.1.1 (long term). *For any subset $F \subset \mathbb{Z}$, does finiteness of all columns of the Serre–Hochschild plane in that range imply smoothness? The cases $F = \{0\}$ and $F = \{1\}$ recover the cohomological and homological questions, while $F = \{0, 1\}$ is the simultaneous question above.*

If every subset admits singular counterexamples, identify a global feature of the full plane that distinguishes smooth algebras from singular ones. For example, this could be its reflection symmetries, multiplication, support, or asymptotic growth, ...

There is also a different intermediate test suggested by the structure of the Serre–Hochschild plane. If A is Gorenstein, then $\omega = DA$ is a 2-sided tilting complex, so the different columns are related by transport along powers of the Serre bimodule. This property alone cannot characterize smoothness. More importantly, Armenta's plane is defined for every finite-dimensional algebra, so the reflection identity below is meaningful without assuming that ω is invertible. For a smooth proper algebra, Armenta proves the stronger Serre reflection

$$\mathbb{T}^{p,m}(A) \cong D \mathbb{T}^{-p, 2-m}(A).$$

Question 1.4. *Let A be a finite-dimensional algebra. If its Serre–Hochschild plane satisfies the smooth-proper Serre reflection above, must A be smooth?*

1.2. Precursor: DG version of Han's conjecture

Smooth proper DG categories are noncommutative analogues of smooth proper varieties. In the DG setting, smoothness means that the diagonal bimodule is perfect, while properness means that morphism spaces of any 2 objects have finite-dimensional cohomology [46, 83]. In particular, smooth finite dimensional algebras and smooth and proper. This makes DG algebras a useful place to test questions of ordinary algebras, while the examples are more abundant. Our DG counterexample came first and provided the expectation that the classical Han's conjecture should be false.

Theorem 1.5 ([64]). *Over every field k of characteristic zero, there exists a finite-dimensional DG k -algebra B which is not smooth but for which $\mathrm{HH}_*(B)$ is finite-dimensional.*

The construction starts from the first Weyl algebra

$$A_1 = k\langle x, y \rangle / (xy - yx - 1),$$

which is smooth but not proper and has finite-dimensional Hochschild homology. We then compactify it categorically. We develop a similar but different notion from Gröbner basis that behaves well with differential, A_1 is realized as a finite-cell DG algebra. Then, Efimov's categorical compactification realizes $\mathrm{Perf}(A_1)$ as a quotient of a proper category with a full exceptional collection. Its kernel is proper but cannot be smooth, while Hochschild localization keeps the Hochschild homology finite-dimensional. Geometric realization and Morita theory then give a finite-dimensional DG algebra model [73].

The DG formulation of the conjecture suggests a geometric question. Although for general proper DG algebras the conjecture can fail, an extra geometry structure can also impose strong additional restrictions: see for example our Cartan determinant result below [66]. In the commutative finite-type setting, eventual vanishing of Hochschild homology forces smoothness [7]; the corresponding global theory for singular schemes is developed in [19]. In particular, basic hypersurface singularities such as nodes and cusps already have nonzero Hochschild homology in arbitrarily high degrees. Thus it is not clear that the DG pathology above can occur for a proper singular variety.

Question 1.6. *Is there a proper singular variety X with finite $\mathrm{HH}_*(X)$?*

The comparison between the DG and ordinary counterexamples suggests a broader issue. Being derived Morita equivalent to an ordinary algebra is extra structure on a DG category, but it is not obvious which homological invariants detect it. Han’s conjecture shows that this extra structure does not restore Hochschild-homological detection of smoothness. The Cartan problem below [66] points in the opposite direction: ordinary smooth finite-dimensional algebras satisfy stronger numerical restrictions than general smooth proper DG categories.

Program 1.2.1 (long term). *Find Morita-invariant homological constraints that distinguish smooth proper DG categories admitting an ordinary finite-dimensional algebra model from genuinely DG ones. Natural tests include Euler and Cartan forms, Hochschild invariants, and realizability of exceptional numerical classes. The goal is to identify statements that make sense for all DG categories but become strictly stronger under ordinary realizability.*

1.3. Numerical Cartan determinants

The failure of smoothness detection by Hochschild (co)homology suggests a different kind of question. Instead of asking whether an invariant vanishes, one can ask what numerical restrictions smoothness and properness impose. The Euler pairing M_χ gives one such restriction. Its determinant $\det(M_\chi)$ extends the Cartan determinant conjecture in classical representation theory to smooth proper DG categories.

The classical Cartan determinant conjecture is an old problem about algebras of finite global dimension. Finite global dimension forces the Cartan determinant to be ± 1 [32]. Zacharia’s 1983 conjecture asks whether -1 can occur [88], and this is often referred as the *Cartan determinant conjecture*. It is known positively for many special families, including quasihereditary-type algebras and broad classes carrying positive or monoid gradings [23, 87, 77], but it remains open in general. Recent work continues to enlarge the known range, for example to representation-finite algebras [33].

For a smooth proper DG category $\mathcal{D}$, there is a natural generalization of this conjecture. The Euler pairing can be degenerate on $K_0(\mathcal{D})$. However, modulo the degenerate locus the right lattice is *numerical Grothendieck group*

$$K_{\text{num}}(\mathcal{D}) = K_0(\mathcal{D})/\ker \chi_{\mathcal{D}}.$$

There, the induced Euler pairing M_χ is nondegenerate. Its determinant is therefore a basis-independent integer, which I call the *numerical Cartan determinant*. For a general smooth proper category, one should not expect $\det(M_\chi) \in \{\pm 1\}$. Geometric examples such as K3 surfaces already give other values. However, to the best of my knowledge, it appears that for every known smooth proper DG category $\mathcal{D}$ in the literature with $\text{rk} K_{\text{num}}(\mathcal{D}) > 0$, we always have $\det(M_\chi) > 0$. This naturally leads to a broader positivity question.

Question 1.7. *Is the numerical Cartan determinant positive for every (non-phantom) smooth proper DG category?*

Positive results for the classical conjecture often use extra algebraic structure. In [87] and [22] the determinant $+1$ statements are proved in positively graded and zero-relations settings, while [77] extends this approach using suitable monoid gradings. Other positive cases, such as quasihereditary-type and representation-finite algebras, use different representation-theoretic structures [23, 33]. Our proof is different: the additional structure is geometric, coming from intersection pairings and Hodge-type positivity rather than from a grading on the algebra.

Theorem 1.8 ([66]). *Let X be a smooth projective variety over an algebraically closed field. Then*

$$\det M_\chi(\text{Perf}(X)) > 0$$

if either $\dim X$ is odd, or $\dim X$ is even and X satisfies the numerical Hodge standard conjecture.

The proof uses the additional geometric structure in an essential way. An abstract smooth proper category has no intersection form on algebraic cycles, but $\text{Perf}(X)$ does. Hirzebruch–Riemann–Roch converts the Euler determinant into intersection pairings on numerical algebraic cycles. In odd dimension, complementary codimensions pair with each other. In even dimension, the middle-dimensional pairing also enters. The numerical Hodge standard conjecture controls its sign. Given that graded-ness and existence of a geometric structure seem very different, it is reasonable to suspect the conjecture actually hold in general. A very fascinating question is to study if the sign of Cartan determinant reveals any potential deeper structure.

Program 1.3.1 (long term). *Explore whether the sign of the numerical Cartan determinant comes from an intrinsic Hodge-type, polarization, or positivity structure on a smooth proper category. In such a theory a negative determinant would signal not merely failure of a numerical conjecture, but failure of a deeper structure that geometry automatically supplies.*

A smooth proper Deligne–Mumford stack is in a natural position between varieties and arbitrary DG categories. Orbifold Riemann–Roch for DM stacks computes the Euler characteristic on the inertia stack, using representation-theoretic Chern characters and twisted-sector terms [82, 27]. The variety argument does not carry over directly. The sectors are paired by the inertia involution and weighted by stabilizer eigenvalues. However, root stacks themselves are not a genuinely new test: their standard semiorthogonal decomposition has one component from the base and the remaining components from the divisor [42]. The Euler matrix is block upper triangular, so positivity follows formally once it is known for those geometric components. The interesting stack cases are therefore those whose derived categories do not split into familiar positive blocks. Another variant comes from Brauer twists. In general, the twisted category of a DM stack $\mathcal{X}$ can have very different behavior from that of $\mathcal{X}$ (see, for example our work [62]). These are summarized into the following program.

Program 1.3.2. *Study Cartan determinant on smooth proper Deligne–Mumford stacks whose derived categories do not admit semiorthogonal decompositions into familiar geometric blocks. For a Brauer class, compare Cartan determinant of the twisted category with the original category. A concrete starting example would be a maximal order on smooth projective surface.*

The Deligne–Mumford stack problem keeps an absolute integer-valued determinant while weakening the geometric setting. A different extension is to keep a geometric base and enrich the value of the Euler pairing itself. Let $\mathcal{A}$ be a sheaf of smooth proper DG algebras over a smooth proper base S , which may be a variety or a suitable DM stack. For perfect $\mathcal{A}$ -modules E, F , consider the relative Euler class

$$\chi_{\mathcal{A}/S}(E, F) = [\mathbf{R}\mathcal{H}om_{\mathcal{A}}(E, F)] \in K_0(S).$$

If the relevant relative numerical group is free of finite rank over $K_0(S)$, a basis gives a Cartan matrix $M_{\chi_{\mathcal{A}/S}}$ with entries in $K_0(S)$ and hence a determinant in that ring. When $S = BG$, this becomes a representation-valued determinant in $R(G)$. This puts families of DG categories and equivariant DG categories into the same framework. The first question is what smoothness and properness can force before one takes dimensions.

Program 1.3.3. *What restrictions do smoothness and properness impose on a relative Cartan determinant in $K_0(S)$?*

- *In the equivariant case $S = BG$, can the determinant be a nontrivial representation class?*
- *For a variety S , there are examples where $\det M_{\chi_{\mathcal{A}/S}} \neq [O_S] \in K_0(S)$. What classes can occur?*

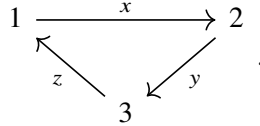
1.4. Finite-cell DG algebras and combinatorial properness

The questions above require explicit noncommutative models where smoothness, properness, and numerical invariants can all be computed. Finite-cell DG algebras provide one such class: its smoothness is automatic, and properness is a relatively combinatorial problem.

A *finite-cell DG quiver* has finitely many vertices and finitely many homogeneous arrows $a_1, \dots, a_N$. They are *ordered*, so that the differential da_i only involve the earlier arrows $a_1, \dots, a_{i-1}$. Ordinary finite quivers are the special case with zero differential, so finite-cell DG quivers are a natural DG generalization of ordinary quivers. Although the presentation is finite, its path algebra can still contains words of arbitrary length, so even a very small DG quiver can carry rich combinatorics. Unlike ordinary quivers, proper finite cell DG quivers can have oriented cycles. For example:

$$z \begin{array}{c} \curvearrowright \\ \curvearrowleft \end{array} 1 \begin{array}{c} \xrightarrow{x} \\ \xleftarrow{y} \end{array} 2, \quad dz = xy.$$

In classical representation theory, one often starts with a finite dimensional algebra and studies its smoothness. Here our problem is different. For a finite-cell DG algebra, smoothness is automatic. *Properness* is the interesting part: it asks finiteness of cohomology of the path DG algebra. It has a combinatorial difficulty: an infinite collection of paths must be controlled by finitely presented differential data. Anick resolutions and algebraic discrete Morse theory provide useful precedents for exactly this kind of compression [1, 80, 45]. They do not solve our problem directly, but they suggest how infinite path combinatorics might be reduced to finite states or critical patterns. A natural question is to find sufficient conditions for properness. In my current work, I obtained several successful examples by using certain combinatorial methods to organize the differentials. For example, fix the base quiver



We attach arrows with differentials, in the following order:

$$\begin{aligned} du_{xy} &= xy, & du_{yz} &= yz, & du_{zx} &= zx, \\ du_{(xyz)} &= m_3(x, y, z), & du_{(yzx)} &= m_3(y, z, x), \\ du_{(xzy)} &= m_4(x, y, z, x). \end{aligned}$$

It can be shown that this DG quiver is proper. It admits a full exceptional collection of length 3, but the combinatorial can be generalized to produce more nontrivial examples.

Algebraic geometry already gives many smooth proper categories. Finite-cell presentations offer a different way to build them. Gradings, filtrations, and local rewriting data can be varied in ways that have no obvious geometric analogue. The compactification picture used in our work [64] also places interesting DG categories as residual pieces of ambient categories with full exceptional collections. The most interesting case is where the category itself and residual category both do not have full exceptional collections.

Program 1.4.1. *Find a sufficient condition (construction method) of properness. I start with low-vertex families and known examples in the literature, obtain their finite-cell DG quiver presentation, and study their combinatorial pattern. Then this can be generalized to more examples that may not occur in the literature.*

A very interesting long term goal is to search for a proper finite-cell DG quiver with negative Cartan determinant.

2. Derived categories and phantom categories

Semiorthogonal decompositions split a derived category into smaller pieces [13]. If we have

$$\text{Perf}(X) = \langle \mathcal{A}_1, \dots, \mathcal{A}_m \rangle,$$

then basic additive invariants split accordingly. In particular, both K_0 and Hochschild homology measure the contributions of the individual components [51]. This suggests a basic detection question: if an admissible component has

$$K_0(\mathcal{A}) = 0, \quad \text{HH}_*(\mathcal{A}) = 0,$$

must the component itself be zero? Surprisingly, the answer is no. A nonzero admissible category with these invariants vanish is called a *phantom category*. The first examples were discovered from surfaces of general type [36, 12]; later constructions showed that the phenomenon also occurs on rational surfaces and in large families [49, 67]. Phantoms are not only limits of additive and motivic invariants: they are often also useful as responsible sources to construct other phenomena. For example, in our work on Han's conjecture [48], a phantom component is eventually converted into a nonzero singularity category with vanishing Hochschild homology.

2.1. Infinitely many phantoms from a 3-vertex quiver

Phantom categories were first found on surfaces of general type, where maximal-length exceptional collections can leave nonzero residual pieces with vanishing K_0 and Hochschild homology [36, 12]. Krah later found a

rational-surface example [49]. Recent work even gives infinite families on one fixed surface [67]. In most geometric examples, however, the phantom itself is hard to study. Its standard additive invariants vanish, while the ambient surface and exceptional collection are large. In our work [65], we take a different route: instead of varying geometric exceptional collections on a surface, we fix one small underlying algebra and obtain infinitely many phantom residuals by changing only a compatible internal grading.

Theorem 2.1 ([65]). *For every $m \in \mathbb{Z}$, there is a DG algebra A_m on a three-vertex quiver with an exceptional triple of maximal length $\langle E_m, F_m, G_m \rangle$ and residual category*

$$\mathcal{P}_m = \langle E_m, F_m, G_m \rangle^\perp \subset \text{Perf}(A_m).$$

The construction has three features:

(i) $\mathcal{P}_m$ is nonzero, but

$$K_0(\mathcal{P}_m) = 0, \quad \text{HH}_*(\mathcal{P}_m) = 0;$$

(ii) *the categories $\mathcal{P}_m$ are pairwise non-derived-Morita-equivalent, while all A_m are regradings of the same 25-dimensional underlying algebra;*

(iii) *a single finite-dimensional bigraded DG algebra B models the entire family:*

$$\mathcal{P}_m \simeq_{\text{Morita}} \text{Perf}(T_m B), \quad T_m : (p, w) \mapsto p + 2mw.$$

Our construction starts from observing a classical directed 3-vertex algebra appearing in [15, 52]. It contains an “unexpected” exceptional object which cannot be extended to a full exceptional collection: its orthogonal contains no exceptional objects. We interpret this not as a reason that the mechanism must stop after one object, but as a special obstruction of that particular algebra. Motivated by this observation, we choose the relations of a carefully designed three-vertex model so that the analogous construction continues through three objects and produces an unexpected exceptional collection of maximal length. Its orthogonal is then nonzero while its standard additive invariants vanish, giving the phantom category above.

To produce infinitely many phantoms, a natural attempt is to assign different gradings on the same DG algebra. We add an internal grading compatible with the algebra and the exceptional modules. This gives us a way to regrade the DG algebra and exceptional modules in a compatible way, and the resulting phantoms indeed have distinct Hochschild cohomology.

A similar phenomenon appeared at almost the same time in algebraic geometry. A countably infinite family of pairwise non-equivalent universal phantoms on the $\text{Bl}_{10} \mathbb{P}^2$ is constructed in [67]. Their parameter varies geometric exceptional collections on a fixed surface. Our parameter varies compatible gradings on a fixed 25-dimensional algebra. The two mechanisms are different, but both show that one starting object can produce infinitely many phantom categories with different Hochschild cohomology.

An important advantage of our example is that its ambient quiver is concrete and small. This makes many studies on internal structures become available by direct computations. For example, the general moduli theory of objects in smooth proper DG categories is available through the work of [85], while [55] gives an algebraic moduli theory for suitable complexes on proper spaces. Although the phantoms on rational surfaces are concrete, the ambient category is too big to carry out actual computations. Using our example, we have the following program.

Program 2.1.1. *Use our small 3-vertex phantom models to understand internal structures of phantoms. Study moduli of objects with fixed numerical data and suitable boundedness conditions, and study hearts of bounded t -structures when explicit models are available. In parallel, compute minimal A_∞ and finite-cell models and look for recurring combinatorial patterns that can be reversed into constructions of new smooth finite-dimensional DG algebras.*

Our method can potentially produce even smaller examples by search more carefully on the ambient DG quivers.

Observe that Krah’s construction [49] is structural: there is an involution that explains where the unexpected exceptional collection comes from; while our construction is less structural: the unexpected exceptional collection

comes from numerical exceptional + sufficiently generic coefficients to guarantee relevant maximal rank conditions. Being less structural suggests 2 programs in opposite directions.

First, an obvious immediate question is to understand whether our construction can be made more structural.

Question 2.2. *Can one recognize, directly from a small quiver, when there is an “unexpected” exceptional collection of maximal length? Can this be turned into a systematic construction of phantom residuals rather than an example-by-example search?*

The second question appears more interesting to me. In fact, being less structural here has its advantage: although there are already many examples of phantom categories in the literature, our example suggests that phantoms are even more ubiquitous than we expected. If a phantom can occur on a 3-vertex DG quiver with little patterns, why don’t we expect the unexpected exceptional collection can be made to admit another phantom? Can we obtain a chain of phantoms? Even more excitingly, can we get an *infinite* chain? From the appearance of our example, this is a highly non-obvious question.

Program 2.1.2. *Build more 3-vertex examples and study the 3-vertex model of their unexpected exceptional collections. Can we build a chain of phantoms? Can the chain be made infinite?*

2.2. New phantom categories on rational surfaces

Phantoms on rational surfaces were initially surprising. The first examples occurred on surfaces of general type [36, 12]. By contrast, strong rigidity results show that del Pezzo surfaces have no phantoms, and some special blowups of $\mathbb{P}^2$ also have none [74, 16]. Then [49] found phantoms on $\text{bl}_{10}\mathbb{P}^2$. Rational surfaces then became a natural place to ask when a maximal-length exceptional collection can fail to be full.

Our work [47] show that the phenomenon continues and suggest a broader picture for rational surfaces.

Theorem 2.3 ([47]). *Let X be the blowup of $\mathbb{P}^2$ at 11 points in general position. There is a semiorthogonal decomposition*

$$D^b(X) = \langle C, E_1, \dots, E_{14} \rangle$$

where the E_i are line bundles forming an exceptional collection and C is a universal phantom.

A second construction on the blowup of $\mathbb{F}_2 = \mathbb{P}_{\mathbb{P}^1}(O \oplus O(2))$ at 9 general points gives another universal phantom. In both cases, a lattice symmetry first gives a numerically exceptional collection. Surface cohomology then shows that the collection is actually exceptional, and pseudoheight detects non-fullness [53]. Hochschild cohomology finally shows that the residual categories are not equivalent to Krah’s phantoms.

The next question is not just whether more phantoms exist. Once a surface contains a phantom, further blowups automatically pullback the old copy to the new one. The harder problem is to find a *new* phantom that knows the deformation space of the blown-up surface. For $X_r = \text{Bl}_r \mathbb{P}^2$, we ask whether the geometric deformations $H^1(T_{X_r}) \subset \text{HH}^2(X_r)$ remain visible in the phantom. Although at first I was optimistic about the answer, the 3-vertex examples [48] above suggest that this may be delicate. Even for a very small algebra, the numerical conditions for an unexpected maximal exceptional collection are restrictive. Phantoms do not appear more often just because the category is “more complicated”: there must be some exactness happening inside.

Question 2.4. *For $r > 11$, does $X_r = \text{Bl}_r \mathbb{P}^2$ admit a new phantom Hochschild cohomology detects all first-order deformations of X_r ? If not, how to understand this obstruction properly?*

This question also fits the conjectural picture for blowups of Hirzebruch surfaces in [47]. Phantoms may become more common, while their size and deformation theory can still depend strongly on the numerical form of the exceptional collection.

2.3. Finiteness properties of t-structures

Beyond existence and deformation, a basic internal question remains: if the usual additive invariants vanish, what kind of abelian structure can a phantom still have? Bounded t -structures are one of the fundamental structures of a triangulated category. Introduced by [10], they produce an abelian heart with kernels, cokernels, filtrations,

and torsion pairs; they are also the starting point for Bridgeland stability conditions and the corresponding moduli theory [17]. The nonexistence problem is highly nontrivial. Compared with the large literature constructing and classifying t -structures, nonexistence theorems are much less common and usually require subtle obstructions. In [3], negative K -theory obstructs bounded t -structures, while [81, 69] show that singularity can obstruct bounded t -structures on $\text{Perf}(X)$.

On phantoms, this question is particularly interesting. Since $K_0 = 0$, a Bridgeland stability central charge cannot exist, but reasonable bounded t -structures may still exist as a substitute [17]. A natural question is therefore whether a phantom can have a bounded heart with reasonable finiteness condition such as Noetherianity or Artinianity. My work [60] shows that this is impossible.

Theorem 2.5 ([60]). *Phantom and quasi-phantom categories cannot admit Noetherian or Artinian bounded t -structures.*

The proof uses a structure of abelian categories called *Krull–Gabriel filtration* on the heart. If the heart of a bounded t -structure is Noetherian or Artinian, its Krull–Gabriel dimension is defined. A classical generator of the triangulated category gives, through its cohomology objects, a generator of the heart $\mathcal{A}$ as a Serre subcategory. This forces the top level of the Krull–Gabriel filtration (i.e. $\mathcal{A}$) to have a predecessor $\mathcal{B}$. The Serre quotient $\mathcal{A}/\mathcal{B}$ is then a nonzero length category, so its $K_0(\mathcal{A}/\mathcal{B})$ nonzero and torsion-free. But it is a quotient of $K_0(\mathcal{A})$, which is torsion or zero. This gives a contradiction.

Recent work [54] gives a result in another direction. They construct bounded t -structures on almost all known phantom and quasiphantom categories, including Krah-type examples, by inducing t -structures on semiorthogonal components. The same method is used in [67] for the new infinite family on $\text{Bl}_{10} \mathbb{P}^2$. By the theorem above, none of these bounded hearts are Noetherian or Artinian. The next problem is therefore to understand what these hearts actually look like.

Program 2.3.1. *Find more concrete models to study these bounded phantom hearts in more detail: study their simple or finitely generated objects, Serre subcategories, and Krull–Gabriel filtrations... A t -structure on a fixed phantom does not itself create a new category, but operations on abelian categories may eventually produce new interesting DG categories.*

3. Derived categories of stacks

Algebraic stacks arise naturally when geometric objects have automorphisms, especially in moduli problems. A quotient stack $[X/G]$ keeps the G -action instead of forgetting it in the orbit space. Its sheaf theory therefore combines geometry with representation theory of stabilizers [72, 70]. Deligne–Mumford stacks are close to ordinary geometry because their stabilizers are finite. Even so, these representations can change numerical invariants, stability conditions, and derived categories. Root stacks give a controlled example by adding a finite cyclic stabilizer along a divisor [24].

I use this setting to ask two questions. When do stabilizer representations create new directions in Bridgeland stability manifolds? How do Brauer twists change the category up to Morita equivalence [56, 84]?

3.1. Stability conditions on root stacks

Classical slope stability on curves organizes vector bundles into moduli spaces. Bridgeland stability extends this idea from sheaves to objects of a triangulated category, and variation of the stability condition produces wall-crossing phenomena in moduli spaces [17]. On surfaces this has become a central tool connecting derived categories with birational and moduli geometry [18, 9]. Extending this theory to Deligne–Mumford surfaces needs new numerical input. Gieseker-type stability and moduli for sheaves on projective DM stacks are developed [70], and [44] proved the Bogomolov–Gieseker inequality in this setting, including root stacks. A root stack still contains more information than its coarse surface: the stacky divisor contributes twisted sectors that ordinary cohomology does not see. The question is whether one can build stability conditions whose deformations keep this full orbifold information.

The theorem of [63] does this by using the larger rational Chen–Ruan cohomology lattice.

Theorem 3.1 ([63]). *Let X be a smooth projective surface over $\mathbb{C}$, let $C \subset X$ be a smooth curve, and let*

$\mathcal{X} = \sqrt[n]{(X, C)}$ be the n -th root stack along C . The tilt stability conditions on $\mathcal{X}$ satisfy the support property with respect to rational Chen–Ruan cohomology.

The proof first establishes the support property for ordinary cohomology. It then controls the orbifold Chern-class directions using subobjects of stable objects. The larger lattice matters because the support property allows the central charge to deform in the new twisted-sector directions. This opens the way to wall-crossing questions that see the stabilizer, not only the coarse surface.

One motivation for this project came from noncommutative surfaces. Moduli theory on maximal orders on surfaces is a developing area [50]. On the stack side, there is also a theory of moduli of semistable and twisted sheaves [70, 56]. A concrete example is a maximal quaternion order on $\mathbb{P}^2$ ramified along a smooth quartic. Its minimal- c_2 A -line bundles form a fine moduli space, and the derived category of that moduli space embeds fully faithfully into the derived category of A -modules [79]. This gives a specific target for extending the root-stack stability theory.

Program 3.1.1. *Use the Bridgeland stability on root-stack as a starting point for stability and wall-crossing on ramified maximal orders, viewed as twisted root stacks.*

3.2. Morita theory and Brauer twists

Morita theory asks when two algebras have the same module category. For Azumaya algebras over a commutative ring, the Brauer group gives the answer. Over schemes, the problem also involves automorphisms of the underlying space and categories of twisted sheaves; reconstruction results make this relation precise [56, 26, 21]. For stacks, stabilizers add extra representation-theoretic data, and even Gabriel-style reconstruction can fail. Root gerbes give a controlled setting in which to ask what extra stacky information the module category remembers. In this case, Morita equivalence depends not only on the Brauer class on the coarse space but also on a finite torsor cover.

Let $n > 0$. Assume that the base field k contains the n -th roots of unity and that $\text{char}(k) \nmid n$. Let X/k be a smooth projective variety, and let $\mathcal{X} \rightarrow X$ be a μ_n -gerbe of a line bundle. A Brauer class on $\mathcal{X}$ can be described by a μ_n -torsor $q : \tilde{X} \rightarrow X$ together with a Brauer class on X . In [62], we give a complete criterion for Morita equivalence. Here, Morita equivalence means equivalence of coherent module categories as k -linear abelian categories.

Theorem 3.2 ([62]). *Let $\mathcal{A}$ and $\mathcal{B}$ be Azumaya algebras on $\mathcal{X}$, with associated μ_n -torsors $q_i : \tilde{X}_i \rightarrow X$ and Brauer classes $[\mathcal{A}'_i] \in \text{Br}(X)$. Then $\mathcal{A}$ and $\mathcal{B}$ are Morita equivalent if and only if there is an isomorphism of algebraic varieties over k*

$$f : \tilde{X}_1 \xrightarrow{\sim} \tilde{X}_2,$$

not necessarily an isomorphism of μ_n -torsors, such that

$$[q_1^* \mathcal{A}'_1] = [f^* q_2^* \mathcal{A}'_2] \quad \text{in } \text{Br}(\tilde{X}_1).$$

Thus Morita equivalence depends on both the Brauer class and the finite cover carried by the stack. An interesting consequence is that Brauer twisting can change decomposability: an untwisted category may decompose while a Brauer twist becomes indecomposable. A related result in our other work [35] shows that the tensor action of $\text{Pic}(X)$ on a twisted derived category is faithful for schemes, while the analogous statement can fail for Deligne–Mumford stacks. Both results show that stabilizers can change categorical structure even when coarse numerical invariants miss this change.

These results suggest a different direction for stacks whose stabilizers are generically trivial. For schemes, Morita theory for twisted sheaves is rigid: the category of twisted quasi-coherent sheaves determines both the underlying scheme and its Brauer class [2]. Related reconstruction and Morita results extend to algebraic spaces and, in a different direction, to $\mathbb{G}_m$ -gerbes [25]. More recently, [43] proved a Bondal–Orlov reconstruction theorem for smooth proper tame algebraic stacks with generically trivial stabilizer, under a natural hypothesis on the canonical bundle. Together, these results suggest that tame Deligne–Mumford stacks with trivial generic stabilizer may retain similar rigidity as that of schemes, in sharp contrast with the gerbe phenomena discussed above.

Program 3.2.1. *Develop a Morita reconstruction theory for Brauer twists of tame Deligne–Mumford stacks with trivial generic stabilizer. Given Brauer classes $\alpha \in \text{Br}(X), \beta \in \text{Br}(\mathcal{Y})$, study if an equivalence*

$$\text{QCoh}(X, \alpha) \simeq \text{QCoh}(\mathcal{Y}, \beta)$$

forces an isomorphism $f: \mathcal{X} \xrightarrow{\sim} \mathcal{Y}$ with $f^*\beta = \alpha$. Starting from $\alpha = \beta = 1$ will generalize the Gabriel reconstruction.

4. Homological methods in vector-bundle geometry

Vector bundles is one of the most fundamental subjects in algebraic geometry, because categories of sheaves remember much information about the underlying variety itself. This philosophy is justified by several reconstruction theorems, such as Gabriel reconstruction from coherent sheaves [34], Bondal-Orlov reconstruction from additive derived categories [14], and Balmer reconstruction from tensor derived categories [8]. My previous study used homological algebra to carry out a sequence of work on rigid vector bundles, such as exceptional bundles on spherical bundles. Their moduli spaces are the simplest, and they contain rich homological information.

4.1. Exceptional bundles on $\mathbb{P}^3$

Exceptional bundles played an important role in the classification of stable bundles on $\mathbb{P}^2$. Rigid stable bundles control the boundary of the stable Chern-character region [31], and exceptional bundles on $\mathbb{P}^2$ are completely classified and generated from line bundles by mutations [76]. In higher dimensions, much less is known. Helix and mutation theory produce many exceptional bundles, and the constructive ones on $\mathbb{P}^3$ have been classified numerically. It is still unknown whether every exceptional bundle on $\mathbb{P}^3$ is constructive [71, 37]. Even deciding whether a given numerical type exists can be difficult. The Euler pairing gives necessary numerical restrictions, but they may miss geometric obstructions. An exceptional bundle E satisfies

$$\mathrm{Hom}(E, E) = \mathbb{C}, \quad \mathrm{Ext}^i(E, E) = 0 \quad (i > 0).$$

Exceptional bundles are rigid and stable, and exceptional collections give some of the most accessible derived categories [76, 71]. Mutation theory on Fano threefolds gives further evidence for this viewpoint [75].

My work shows that the numerical conditions are not sufficient: there are deeper obstructions caused by geometry. If E is exceptional on $\mathbb{P}^3$ with rank r and degree d , restriction to a general quartic K3 surface gives a stable spherical bundle. This implies the necessary divisibility condition

$$r \mid (2d^2 + 1).$$

The theorem in [61] shows that many numerical solutions still do not come from actual exceptional bundles.

Theorem 4.1 ([61]). *On $\mathbb{P}^3$, there is no exceptional bundle of rank $2d^2 + 1$ and degree d for $|d| \geq 4$.*

The proof adds geometry to a problem that is hard to see from the exceptional equations alone. We use a pencil of quartic K3 surfaces and study the rigid bundles forced on its members by the numerical data. If a bundle existed on $\mathbb{P}^3$, these rigid restrictions would have to fit together in the family. For the maximal-rank family, they do not descend from the total family to $\mathbb{P}^3$. Furthermore, this is not the only obstruction. We also show the nonexistence result for a numerical data out of this range: rank-27 and degree-11.

The theorem shows that the Euler-form equations can miss a geometric obstruction. Any exceptional object generates a semiorthogonal component equivalent to $\mathrm{Perf}(k)$, so the same restriction method also limits possible exceptional decompositions and their algebra models. The quartic K3 surfaces are therefore more than a technical tool: their explicit stability theory provides a movable geometric test for a hypothetical exceptional object.

This viewpoint also shows exactly what may be lost outside algebraic geometry. In quantum $\mathbb{P}^3$ -type settings, a comparable family of K3 sections may not exist. A similar difficulty appears in the 3-vertex work above [65]. A variant of the current ambient quiver produces another rank-1 residual category not equivalent to $\mathrm{Perf}(k)$ but with the same numerical lattice as $\mathrm{Perf}(k)$. There, existence of a third exceptional object is not clear. The lack of geometric structure makes excluding this possibility extremely difficult. This suggests looking for an algebraic substitute for the geometric restriction test.

Question 4.2. *What algebraic structure can replace restriction to movable K3 sections and obstruct exceptional objects in rank-one residual DG categories that are numerically indistinguishable from $\mathrm{Perf}(k)$? If there is such a method, can it improve our nonexistence result on $\mathbb{P}^3$?*

4.2. Spherical vector bundles on smooth and nodal K3 surfaces

Spherical bundles are special rigid objects on K3 surfaces. Mukai's theory describes moduli of stable sheaves using the Mukai lattice, and spherical classes produce spherical twists and many derived autoequivalences [68, 41]. Bridgeland stability and wall-crossing turn this lattice picture into a practical tool for moduli and birational geometry [18, 9]. Even basic questions such as the cohomology of a stable sheaf on a K3 surface can be hard. Broad analysis for general stable sheaves are carried out [29], while precise cohomology dimensions were not completed. A spherical object is rigid, and further requiring stability makes it unique. Thus the exact cohomology dimension is determined by its numerical data.

In [58], I use Bridgeland wall-crossing to give an algorithm that computes the cohomology of every stable spherical vector bundle from its Mukai vector. When $\text{Pic}(X) = \mathbb{Z}H$, the method also gives an explicit numerical criterion for the vanishing of $H^1(X, E)$.

In [59], I study spherical vector bundles on nodal K3 surfaces. Let X be a nodal K3 surface of Picard rank 1, and let $\tilde{X} \rightarrow X$ be its resolution. By moving the problem to the smooth K3 surface $\tilde{X}$ and studying stability there, I determine existence and uniqueness of stable spherical bundles for positive-rank spherical Mukai vectors. There is one rank-2 exception when the class group is larger than the Picard group. These examples give a controlled setting for comparing rigid objects under resolution and degeneration. Both [61, 59] use the cohomology algorithm previously developed in [58].

From my current perspective, a natural question is whether this rigidity can itself be recognized from an explicit algebraic model of the ambient K3 category. Choose a classical generator G of $\text{Perf}(X)$ and set $A = \mathbf{R}\text{Hom}(G, G)$. A spherical bundle becomes a perfect A -module M with

$$H^* \mathbf{R}\text{Hom}_A(M, M) \cong k \oplus k[-2].$$

The self-endomorphism algebra itself is too small to contain complicated higher structure. There must be deeper explanation of this rigidity: M should be a very special representation of A . Since the set of spherical objects in K3 surfaces of Picard rank ≥ 2 is still mysterious, this approach can be used to study this classification problem. A concrete starting point is studying the minimal A_{inf} model in [78].

Program 4.2.1. *For an explicit DG model A of a K3 category, can spherical bundles be characterized intrinsically as a distinguished class of perfect A -modules? Which representation-theoretic feature of the ambient module structure forces $\text{Ext}_A^1(M, M) = 0$?*

Build explicit DG models of K3 categories in which spherical bundles can be studied as concrete modules, and compare the resulting representation-theoretic structure with their wall-crossing and cohomology behavior.

Rigid bundles describe one extreme of moduli theory, where deformation is strongly constrained. A complementary question is how cohomology behaves in positive-dimensional families of stable bundles. In this direction, I have also worked on higher-rank Brill–Noether theory on $\mathbb{P}^2$ [38]. We found a sharp difference between the degree 1 and larger degrees cases. In the latter case, the loci can be reducible and can have components larger than expected. This work remains part of my broader interest in moduli and wall-crossing.

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